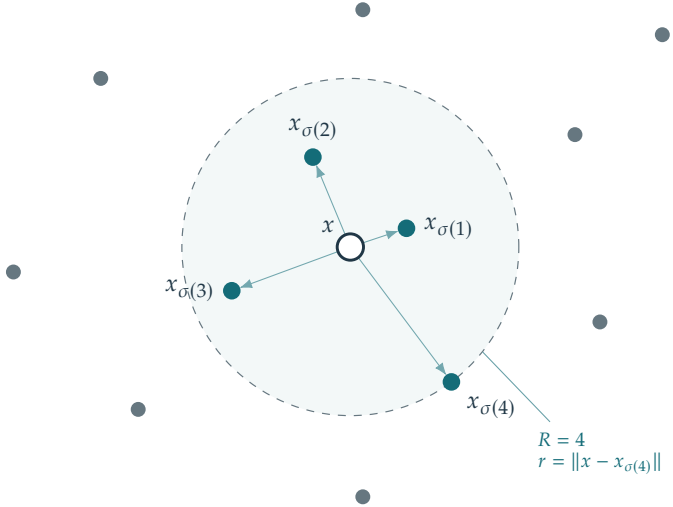

$$\|x - x_{\sigma(1)}\| \leq \cdots \leq \|x - x_{\sigma(4)}\| < \|x - x_{\sigma(5)}\|$$